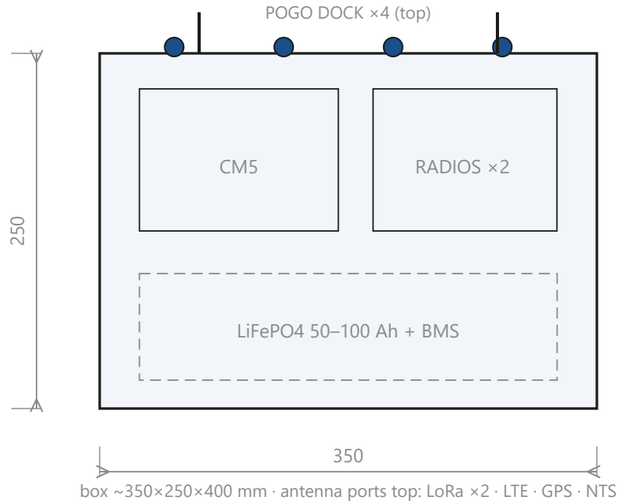
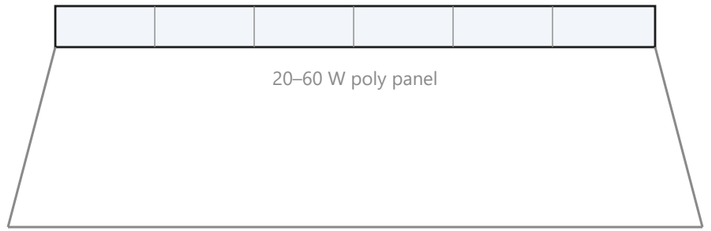
FRONT ELEVATION



box ~350×250×400 mm · antenna ports top: LoRa ×2 · LTE · GPS · NTS

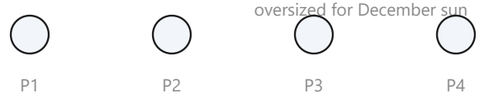
SOLAR ARRAY (SEPARATE)



SWARM DOCK DETAIL

ground frame / pole mount

oversized for December sun



Mite bottom has 2 pogo pads + alignment notch — lands blind, charges, rebroadcasts. Each dock ~5 V / 500 mA via CN3791-class controller.

WHAT THE ANCHOR DOES

FUNCTION	DETAIL
Swarm dock	4 pogo-pin charge positions — Miters land, charge, rebroadcast
Long-range relay	SX1262 at height + bigger antenna — extends mesh over ridges
Gateway	LTE/Wi-Fi/sat backhaul when available — the mesh's door out
Anchor sensor	own temp/RH/VOC + wind — fixed reference for swarm data
Status beacon	E-Ink or LED shows mesh health — readable infrastructure

WHY ANCHORS MATTER

— Docked Miters top off → perpetual patrol without battery swaps.

— Elevated relay roughly doubles effective mesh diameter vs. ground nodes.

— The visible, maintainable heart of the local mesh — trust infrastructure.

— Gateways are what communities pay for — the swarm rides on top.

ARCTICCHAIN — RURAL NODE FLEET
EMEBET INC. · emebetos.ca
SOLE PROPRIETARY OWNER
SURAFEL NEGUSSIE — PLANNER/MAKER
PROPRIETARY & CONFIDENTIAL
UNAUTHORIZED DISTRIBUTION PROHIBITED

DRAWN **Surafel Negussie**
SHEET **1/2**
DATE **2026-11-12**
DWG NO **AF-A-101**
SCALE **NTS**
UNITS **mm**

PARTS LIST — PER STATION

#	PART	SUPPLIER	APPROX CAD	QTY
1	CM5 4GB + IO Board (gateway compute)	PiShop / Canada Computers	$90–130	1
2	SX1262 LoRa breakout ×2 (node + backbone)	DFRobot / AliExpress	$30–50	2
3	LTE HAT OR Wi-Fi bridge OR sat terminal	DFRobot / Starlink / Iridium	$60–600	0–1
4	20–60 W poly panel + MPPT 20–30 A	Renogy / Amazon.ca	$150–400	1
5	LiFePO4 12.8 V 50–100 Ah + BMS	Amazon.ca / supplier	$400–900	1
6	IP65 box ~350×250×400 + pole/frame	Bud Industries / fabricator	$200–400	1
7	Pogo dock ×4 + charge controllers	custom / CN3791	$40–80	4
8	Antennas: LoRa whip ×2, LTE whip, GPS patch	electronics supplier	$60–120	1 set

POWER SYSTEM — SIZED FOR WINTER

20–60 W ARRAY (oversized Dec) → **MPPT 20–30 A** (LiFePO4) → **LiFePO4 50–100 Ah** (640–1280 Wh) → **BMS → 5/12 V** (station + dock)

Autonomy target 30–90 days no-sun — LoRa is µW-class, compute sleeps.

BACKHAUL OPTIONS (HIERARCHY)

OPTION	COST	WHEN
Mesh only	$0	no coverage — still relays + docks
LTE (SIM7600)	$60–90	any cell signal — most rural sites
Starlink Mini/Res	$300–600 + $/mo	true dead zones — business case
Iridium Certus/L-band	OEM $500+	polar/north — own-sat path
ArcticChain sats	future	the mesh IS the ground segment

DOCK PROTOCOL

— Mite lands on pogo pads → charge-detect line pulled low.
— Dock controller broadcasts slot assignment over LoRa.
— Charge until >80% or timeout; release + 'charged' beacon.
— Rotating priority: lowest-battery docked unit gets next slot.
— All dock events signed + logged — swarm energy is mesh data.

FIELD ACCEPTANCE

— 4 Miters dock, charge, undock on schedule — 72 h soak.
— ≥10 signed msgs/min relayed through Anchor at 5 km spacing.
— LTE failover: drop → store-forward continues → auto-reconnect.
— Winter: -30 °C, panel snow-covered 3 days → still docks + relays.

ARCTICCHAIN — RURAL NODE FLEET
EMEBET INC. · emebetos.ca
SOLE PROPRIETARY OWNER
SURAFEL NEGUSSIE — PLANNER/MAKER

PROPRIETARY & CONFIDENTIAL
UNAUTHORIZED DISTRIBUTION PROHIBITED

DRAWN	**Surafel Negussie**
SHEET	**2/2**
DATE	**2026-11-12**
DWG NO	**AF-A-102**
SCALE	**NTS**
UNITS	**mm**